



Mumen Wehbe

Structural Analysis / FEA Engineer

ABOUT ME

Mechanical and mechatronics engineer with dual expertise in physical systems and advanced structural analysis (FEA/IGA). Specialized in numerical modeling, solver validation, and real-time optimization (ROM). Passionate about solving complex physical problems, I am looking for a CDI opportunity as a Structural Analysis / Numerical Simulation Engineer.

CONTACT

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- Paris, France

SKILLS

- C++
- PYTHON
- Matlab
- SolidWorks
- Abaqus
- MSC PATRAN/NASTRAN
- COMSOL
- SAM Modeling
- Web Design
- Microsoft Office

LANGUAGES

- English (Fluent)
- Arabic (Native)
- French (Intermediate)

EDUCATION

- (2024 – 2026)
LE CNAM – PARIS, FRANCE
MASTER'S DEGREE IN MECHANICAL ENGINEERING (M2)
Grade: **A-**
- (2018 – 2022)
CYPRUS INTERNATIONAL UNIVERSITY – NICOSIA, CYPRUS
BACHELOR OF SCIENCE IN MECHATRONICS ENGINEERING (BSC)
Grade: **A**

PROFESSIONAL EXPERIENCE

- (04/2026 – 09/2026)
R&D INTERN – DIGITAL TWIN & ISOGEOMETRIC ANALYSIS (IGA)
LE CNAM (EPN04 LABORATORY) – PARIS, FRANCE
 - Reduced Order Modeling (ROM) via Isogeometric Analysis (IGA) and hyper-reduction (ECSW, DEIM, Gappy POD) of structures in large displacements.
 - Development of a native JS 2D IGA computational core for real-time online resolution (< 10 ms, 80x–120x speedup).
 - Design of an interactive Digital Twin (web 3D Three.js, real-time multiphysics parameters) coupled with a 3D-printed physical demonstrator.
 - Metrological validation and convergence of solvers on reference test cases (Kirsch, non-linear structural beam bending).
- (11/2022 – 02/2023)
ROBOTICS & STEM INSTRUCTOR
STEMA & FUN ROBOTICS – DUBAI, UNITED ARAB EMIRATES
 - Designed and implemented advanced robotics curricula (Vex IQ, Python) and led interactive workshops at the Sharjah International Book Fair.
 - Coached student teams for the Vex V5 competition (achieving prestigious awards in 'Excellence' and 'Design' categories) and provided custom training sessions.
- (07/2022 – 08/2022)
SOLAR ENGINEER
ALEMDAR TEKNIK – NICOSIA, CYPRUS
 - Designed and supervised residential solar PV installations for on-grid and off-grid systems, utilizing SAM modeling for performance analysis.
 - Diagnosed and resolved complex technical issues in photovoltaic systems (inverters, batteries, and panels), ensuring optimal energy yield.